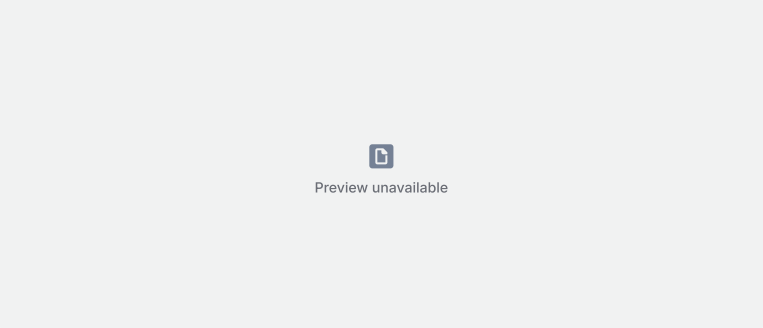
SQL Assesment

Question 1

```sql
USE AdventureWorks2022;

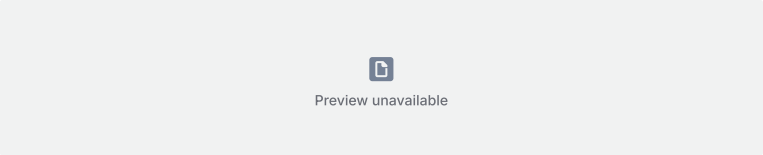
SELECT * FROM HumanResources.Employee
```

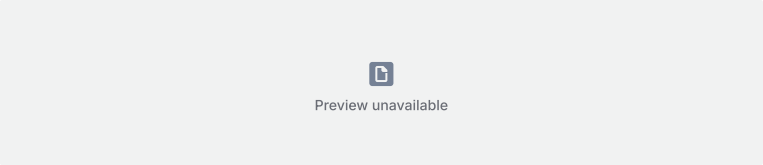


Question 2

```sql
USE AdventureWorks2022;
/*Employee table*/
SELECT * FROM HumanResources.Employee e
LEFT JOIN Person.Person p
ON e.BusinessEntityID = p.BusinessEntityID
ORDER BY p.LastName ASC

/*Person table*/
SELECT * FROM Person.Person p
ORDER BY p.LastName ASC
```

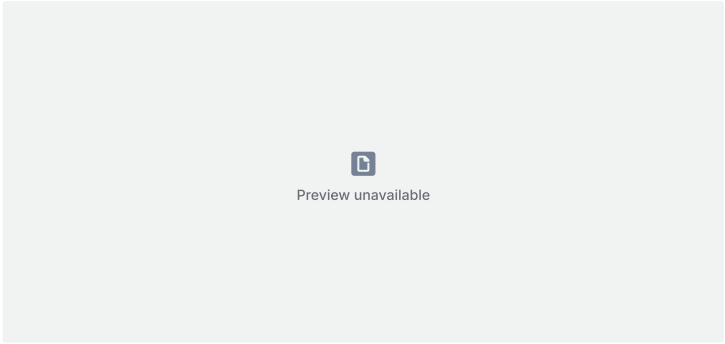




Question 3

```sql
USE AdventureWorks2022;

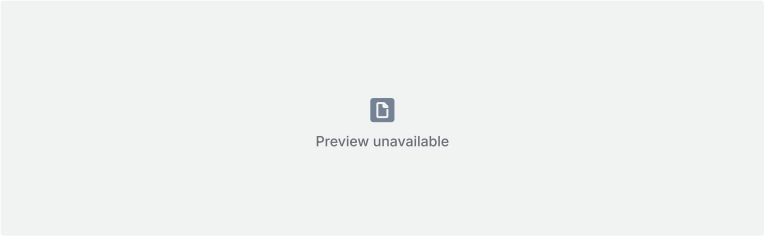
SELECT FirstName, LastName, BusinessEntityID as 'Employee_id'
FROM Person.Person
ORDER BY LastName ASC
```



Question 4

```
USE AdventureWorks2022;

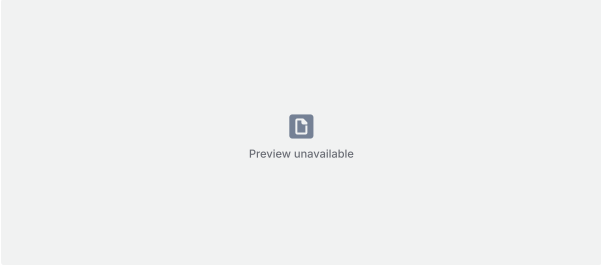
SELECT * FROM Production.Product
WHERE (SellStartDate IS NOT NULL) AND (ProductLine = 'T')
```



Question 5

```
USE AdventureWorks2022;

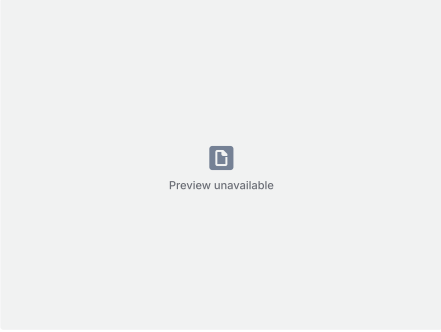
SELECT SalesOrderID,CustomerID,OrderDate, SubTotal,
(TaxAmt/SubTotal)*100 AS 'Tax_percent'
FROM Sales.SalesOrderHeader
ORDER BY SubTotal DESC
```



Question 6

```
USE AdventureWorks2022;

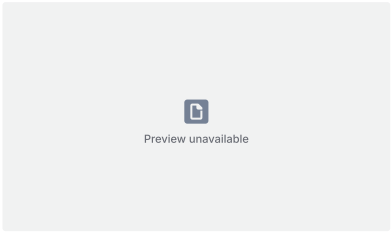
SELECT DISTINCT(JobTitle) FROM HumanResources.Employee
```



Question 7

```
USE AdventureWorks2022;

SELECT CustomerID, SUM(Freight) as 'Total freight'
FROM Sales.SalesOrderHeader
GROUP BY CustomerID
ORDER BY CustomerID
```

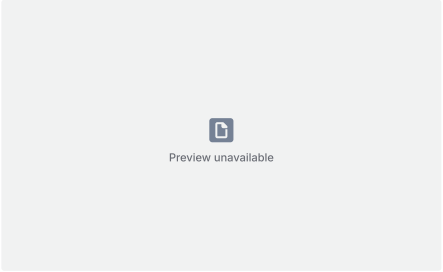


Question 8

```
USE AdventureWorks2022;

SELECT CustomerID, SalesPersonID, AVG(SubTotal) as 'avg_subtotal',
SUM(SubTotal) as 'sum_subtotal'
FROM Sales.SalesOrderHeader
GROUP BY CustomerID, SalesPersonID
ORDER BY CustomerID DESC

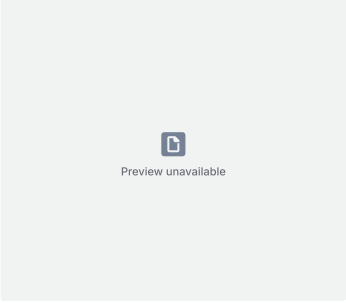
```



Question 9

```
USE AdventureWorks2022;

SELECT ProductID, SUM(Quantity) as 'Total Quantity'
FROM Production.ProductInventory
WHERE (Shelf = 'A'OR Shelf = 'C' OR Shelf = 'H')
GROUP BY ProductID
Having SUM(Quantity)>500
ORDER BY ProductID ASC
```

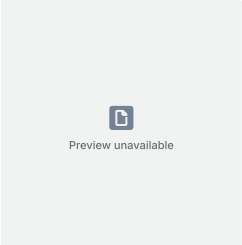


Question 10

```
USE AdventureWorks2022;

SELECT SUM(Quantity) as 'Total Quantity'
FROM Production.ProductInventory
Group by LocationID*10
```

```
6
7
```

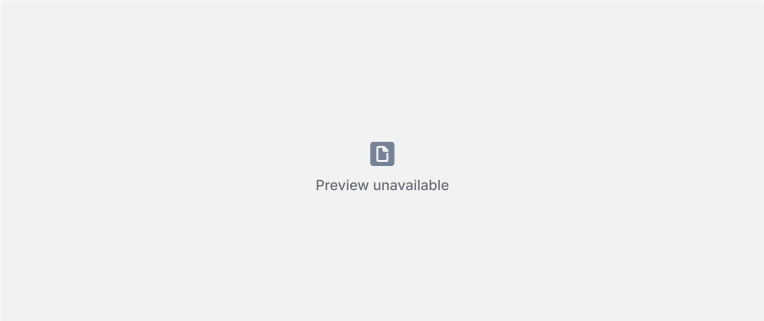


Question 11

```
USE AdventureWorks2022;

SELECT  P.BusinessEntityID, FirstName, LastName, N.PhoneNumber
FROM Person.Person P
LEFT JOIN Person.PersonPhone N
ON p.BusinessEntityID = N.BusinessEntityID
WHERE LastName LIKE 'L%'
ORDER BY  LastName,FirstName


```

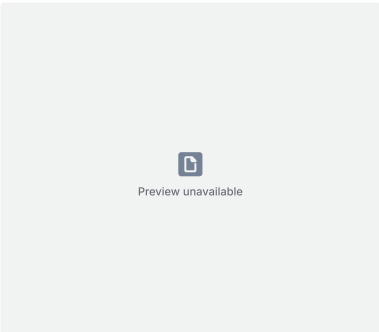


Question 12

```
USE AdventureWorks2022;

SELECT  DISTINCT(SalesPersonID), CustomerID, SUM(SubTotal) AS
'Sum_Total'
FROM Sales.SalesOrderHeader
WHERE SalesPersonID IS NOT NULL
GROUP BY SalesPersonID, CustomerID
ORDER BY SalesPersonID ASC


```



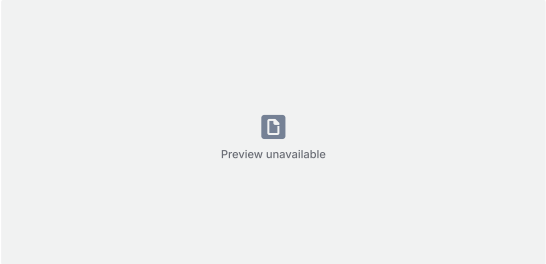
Question 13

```
USE AdventureWorks2022;

SELECT DISTINCT(LocationID), Shelf, SUM(Quantity) AS 'Total Quantity'
FROM Production.ProductInventory
```

```
GROUP BY LocationID, Shelf

```

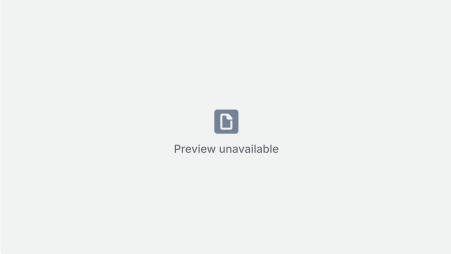


Question 14

```
USE AdventureWorks2022;

SELECT LocationID,Shelf, SUM(Quantity) AS 'TotalQuantity'
FROM Production.ProductInventory
GROUP BY ROLLUP (LocationID, Shelf)
ORDER BY LocationID, Shelf;

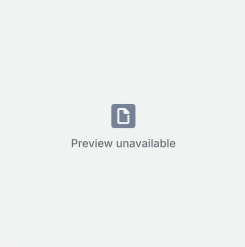

```



Question 15

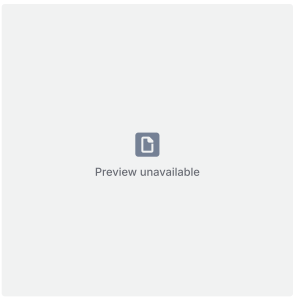
```
USE AdventureWorks2022;

SELECT LocationID, SUM(Quantity) AS 'TotalQuantity'
FROM Production.ProductInventory
GROUP BY ROLLUP (LocationID)
ORDER BY LocationID;
```



Question 16

```
SELECT  City, COUNT(BusinessEntityID) AS 'No. Of Employees'
FROM Person.BusinessEntityAddress b
LEFT JOIN Person.Address a
ON b.AddressID = a.AddressID
GROUP BY City
ORDER BY City ASC
```

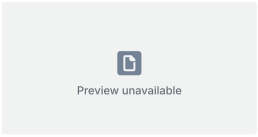


Question 17

```
USE AdventureWorks2022;

SELECT  DATEPART(YYYY,OrderDate) AS 'Year', SUM(TotalDue) AS 'Order Amount'
FROM Sales.SalesOrderHeader
GROUP BY DATEPART(YYYY,OrderDate)
ORDER BY DATEPART(YYYY,OrderDate) ASC

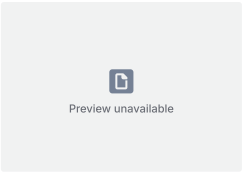

```



Question 18

```
USE AdventureWorks2022;

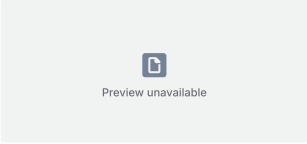
SELECT DATEPART(YYYY,OrderDate) AS 'Year', SUM(TotalDue) AS 'Order Amount'
FROM Sales.SalesOrderHeader
WHERE OrderDate<=2016
GROUP BY DATEPART(YYYY,OrderDate)
ORDER BY DATEPART(YYYY,OrderDate) ASC
```



Question 19

```
USE AdventureWorks2022;

SELECT  ContactTypeID, Name
FROM Person.ContactType
WHERE Name LIKE '%Manager'
```



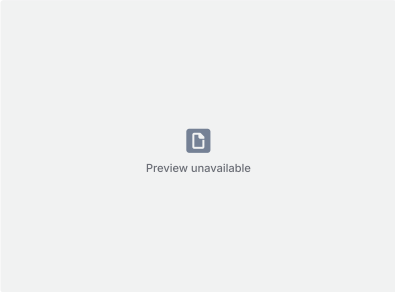
Question 20

```
SELECT pp.FirstName,pp.LastName, bc.BusinessEntityID
FROM Person.BusinessEntityContact bc
LEFT JOIN Person.ContactType ct
ON bc.ContactTypeID = ct.ContactTypeID
LEFT JOIN Person.Person pp
ON  bc.BusinessEntityID = pp.BusinessEntityID
WHERE ct.Name = 'Purchasing Manager'
ORDER BY LastName,FirstName
```

ERROR WILL COME BACK

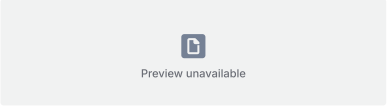
QUESTION 21

```
SELECT ROW_NUMBER() OVER(PARTITION BY pa.PostalCode ORDER BY
sp.SalesYTD DESC) AS 'Row Num',  pp.LastName, sp.SalesYTD,
pa.PostalCode
FROM Person.BusinessEntityAddress ba
INNER JOIN Person.Address pa
ON ba.AddressID = pa.AddressID
INNER JOIN Sales.SalesPerson sp
ON ba.BusinessEntityID = sp.BusinessEntityID
INNER JOIN Person.Person pp
ON ba.BusinessEntityID = pp.BusinessEntityID
ORDER BY sp.SalesYTD DESC, pa.PostalCode ASC
```



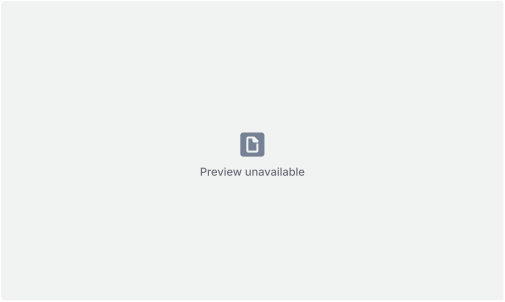
Question 22

```
SELECT  ct.ContactTypeID, ct.Name, COUNT(ct.ContactTypeID) AS 'Number
of Contacts'
FROM Person.BusinessEntityContact bc
LEFT JOIN Person.ContactType ct
ON bc.ContactTypeID = ct.ContactTypeID
GROUP BY ct.ContactTypeID, ct.Name
HAVING  COUNT(ct.ContactTypeID) >= 100
ORDER BY COUNT(ct.ContactTypeID) DESC
```



Question 23

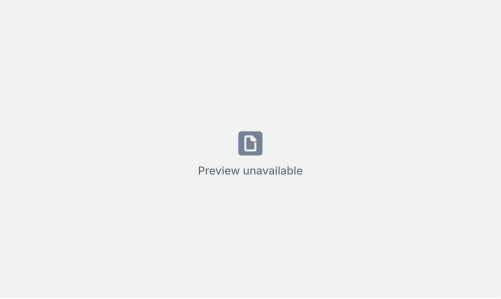
```
SELECT FORMAT( RateChangeDate,'yyyy-MM-dd') AS 'fromdate',
(FirstName+' '  + LastName +' ' + ','+MiddleName) AS 'Fullname', Rate * 40 AS 'Weekly
rate'
FROM HumanResources.EmployeePayHistory ep
INNER JOIN Person.Person pp
ON ep.BusinessEntityID = pp.BusinessEntityID
WHERE FirstName IS NOT NULL AND LastName IS NOT NULL AND MiddleName IS NOT NULL
ORDER BY 'Fullname' ASC
```



Question 24

```
SELECT *
```

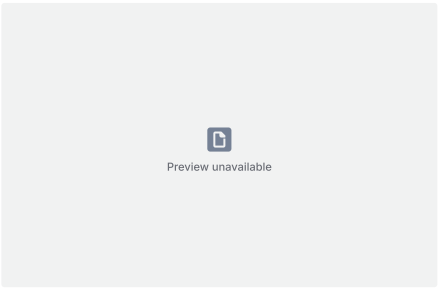
```
FROM (SELECT ROW_NUMBER() OVER( PARTITION BY (FirstName+' '  +
LastName +' ' + ','+MiddleName)  ORDER BY FORMAT(
RateChangeDate,'yyyy-MM-dd')DESC) AS Rowno,
FORMAT( RateChangeDate,'yyyy-MM-dd') AS fromdate,
(FirstName+' '  + LastName +' ' + ','+MiddleName) AS Fullname,
Rate * 40 AS 'Weekly rate'
FROM HumanResources.EmployeePayHistory ep
INNER JOIN Person.Person pp
ON ep.BusinessEntityID = pp.BusinessEntityID
WHERE FirstName IS NOT NULL AND LastName IS NOT NULL AND MiddleName IS
NOT NULL
) AS tbl
WHERE Rowno =1
ORDER BY Fullname
```



Question 25

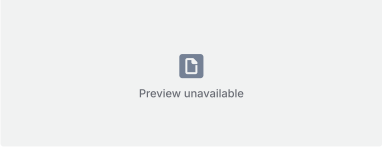
```
SELECT SalesOrderID,ProductID,
OrderQty,
SUM(OrderQty) OVER(PARTITION BY SalesOrderID)AS 'Sum',
COUNT(OrderQty)OVER(PARTITION BY SalesOrderID) AS'Total Qunatity',
AVG(OrderQty) OVER(PARTITION BY SalesOrderID) AS 'Average',
MIN(OrderQty) OVER(PARTITION BY SalesOrderID) AS 'Min',
MAX(OrderQty) OVER(PARTITION BY SalesOrderID) AS 'Max'
FROM Sales.SalesOrderDetail
WHERE SalesOrderID = 43659 OR SalesOrderID = 43664
GROUP BY SalesOrderID,ProductID, OrderQty

```



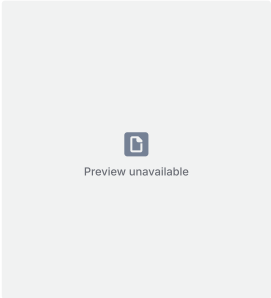
Question 26

```
SELECT SalesOrderID,ProductID,
OrderQty,
SUM(OrderQty) OVER(PARTITION BY SalesOrderID)AS 'Sum',
COUNT(OrderQty)OVER(PARTITION BY SalesOrderID) AS'Total Qunatity',
AVG(OrderQty) OVER(PARTITION BY SalesOrderID) AS 'Average'
FROM Sales.SalesOrderDetail
WHERE (SalesOrderID = 43659 OR SalesOrderID = 43664 ) AND
CAST(ProductID AS VARCHAR(10)) LIKE '71%'
GROUP BY SalesOrderID,ProductID, OrderQty
```



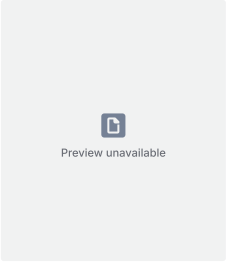
Question 27

```sql
SELECT SalesOrderID, SUM(LineTotal) AS OrderIDTotal
FROM Sales.SalesOrderDetail
GROUP BY SalesOrderID
HAVING SUM(LineTotal) > 100000
```



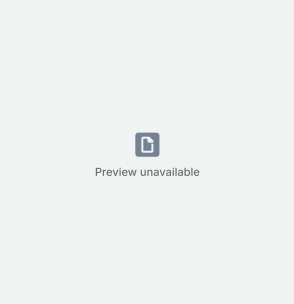
Question 28

```sql
SELECT ProductID, Name
FROM Production.Product pp
WHERE pp.Name  LIKE 'Lock Washer%'
```



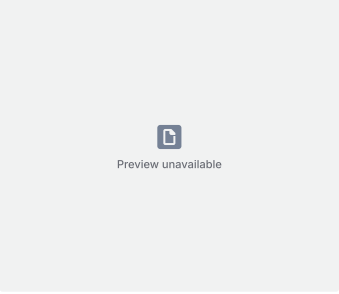
Question 29

```sql
SELECT ProductID, Name,Color
FROM Production.Product
ORDER BY ListPrice ASC
```



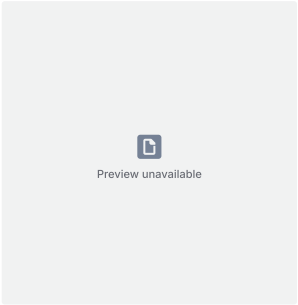
Question 30

```sql
SELECT BusinessEntityID, JobTitle,HireDate
FROM HumanResources.Employee
ORDER BY HireDate
```



Question 31

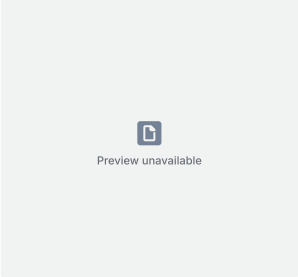
```sql
SELECT LastName,FirstName
FROM Person.Person
WHERE LastName LIKE 'R%'
ORDER BY FirstName ASC, LastName DESC
```



Question 32

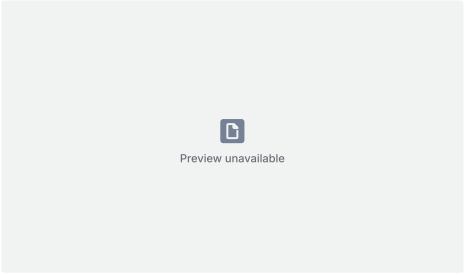
```sql
SELECT BusinessEntityID,
    IIF (SalariedFlag = 1, 'true', 'false') AS 'SalariedFlag'
FROM HumanResources.Employee
ORDER BY IIF (SalariedFlag =  1 ,BusinessEntityID  ,-BusinessEntityID )

```



Question 33

```sql
SELECT ss.BusinessEntityID,pp.LastName, ss.TerritoryID, st.Name,cr.Name
FROM Sales.SalesPerson ss
INNER JOIN Person.Person pp
ON ss.BusinessEntityID = pp.BusinessEntityID
INNER JOIN Sales.SalesTerritory st
on ss.TerritoryID = st.TerritoryID
INNER JOIN Person.CountryRegion cr
ON st.CountryRegionCode = cr.CountryRegionCode
ORDER BY IIF(cr.Name LIKE 'United States',st.Name,cr.Name)
```

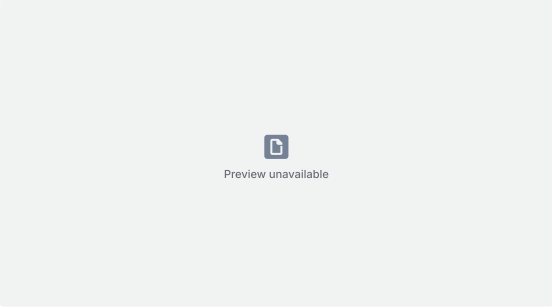


Question 34

```sql

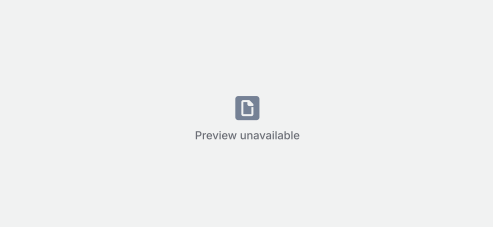
SELECT pp.FirstName,pp.LastName,
    ROW_NUMBER() OVER(ORDER BY pa.PostalCode) AS 'Row Number',
    RANK() OVER(ORDER BY pa.PostalCode) AS 'Rank',
    DENSE_RANK() OVER(ORDER BY pa.PostalCode) AS 'Dense Rank',
    NTILE(4) OVER(ORDER BY pa.PostalCode) AS 'Quartile',
    ss.SalesYTD,
```

```
    pa.PostalCode
FROM Sales.SalesPerson ss
INNER JOIN Person.Person pp
ON ss.BusinessEntityID = pp.BusinessEntityID
INNER JOIN Person.BusinessEntityAddress ba
ON ss.BusinessEntityID = ba.BusinessEntityID
INNER JOIN Person.Address pa
ON ba.AddressID = pa.AddressID
ORDER BY PostalCode
```



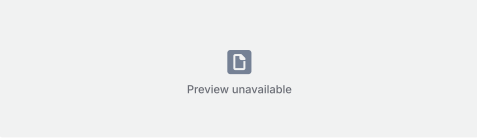
Question 35

```
SELECT DepartmentID,Name,GroupName
FROM HumanResources.Department d
ORDER BY DepartmentID
OFFSET 5 ROWS
```



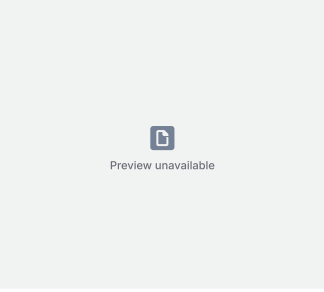
Question 36

```
SELECT DepartmentID,Name,GroupName
FROM HumanResources.Department d
ORDER BY DepartmentID
OFFSET 5 ROWS
FETCH NEXT 5 ROWS ONLY
```



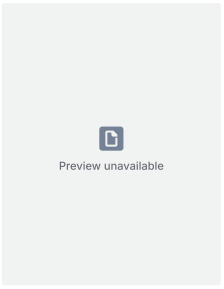
Question 37

```
SELECT Name,Color, ListPrice
FROM Production.Product
WHERE Color = 'Red' OR Color = 'Blue'
ORDER BY ListPrice ASC
```



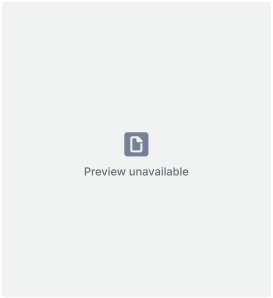
Question 38

```
SELECT pp.Name,sd.SalesOrderID
FROM Sales.SalesOrderDetail sd
FULL JOIN Production.Product pp
ON sd.ProductID = pp.ProductID
ORDER BY pp.Name ASC
```



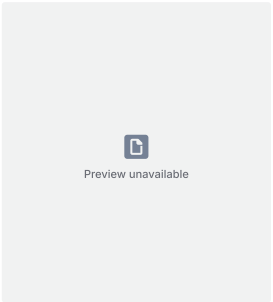
Question 39

```
SELECT pp.Name,sd.SalesOrderID
FROM Sales.SalesOrderDetail sd
RIGHT JOIN Production.Product pp
ON sd.ProductID = pp.ProductID
ORDER BY pp.Name ASC
```



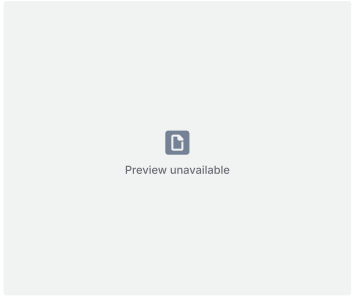
Question 40

```
SELECT pp.Name,sd.SalesOrderID
FROM Sales.SalesOrderDetail sd
INNER JOIN Production.Product pp
ON sd.ProductID = pp.ProductID
ORDER BY pp.Name ASC
```



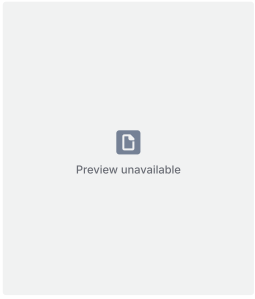
Question 41

```
SELECT st.Name AS 'Territory',sp.BusinessEntityID
FROM Sales.SalesPerson sp
LEFT JOIN Sales.SalesTerritory st
ON sp.TerritoryID = st.TerritoryID
```



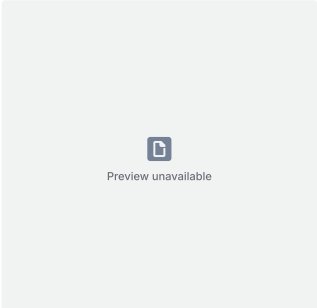
Question 42

```
SELECT FirstName + ' '+ LastName  AS FullName, pa.City
FROM Person.Person pp
JOIN Person.BusinessEntityAddress ba
ON pp.BusinessEntityID = ba.BusinessEntityID
JOIN Person.Address pa
ON ba.AddressID = pa.AddressID
ORDER BY LastName, FirstName
```



Question 43

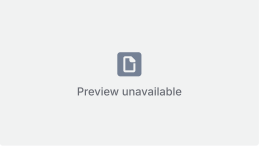
```
SELECT BusinessEntityID, FirstName,LastName
FROM (SELECT*
FROM Person.Person
WHERE LastName LIKE 'Adams' ) AS dt
WHERE PersonType LIKE 'IN'
ORDER BY FirstName
```



Question 44

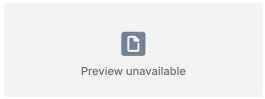
```
SELECT BusinessEntityID, FirstName,LastName
FROM (SELECT*
FROM Person.Person) AS dt
WHERE (BusinessEntityID < 1500) AND (LastName LIKE 'AL%') AND
(FirstName LIKE 'M%')

```



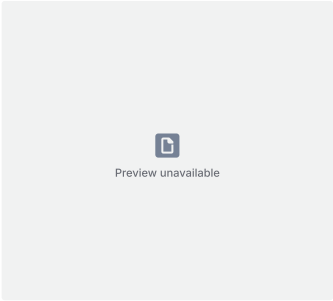
Question 45

```
SELECT ProductID, Name, Color
FROM (SELECT*
FROM Production.Product) AS dt
WHERE Name IN('Blade','Crown Race','AWC Logo Cap')
```



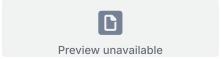
Question 46

```
SELECT SalesPersonID,
    COUNT(SalesPersonID) AS TotalSales,
    YEAR(OrderDate) AS SalesYear
FROM Sales.SalesOrderHeader
WHERE SalesPersonID IS NOT NULL
GROUP BY SalesPersonID, YEAR(OrderDate)
ORDER BY SalesPersonID, YEAR(OrderDate)
```



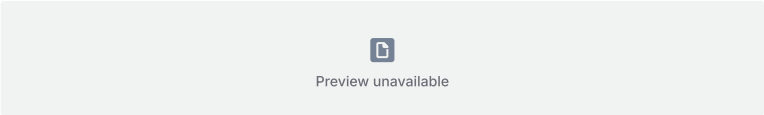
Question 47

```
SELECT SUM(TotalSales)/COUNT(DISTINCT SalesPersonID) AS ' Average Sales Per Person'
FROM (SELECT SalesPersonID,
    COUNT(SalesPersonID) AS TotalSales,
    YEAR(OrderDate) AS SalesYear
FROM Sales.SalesOrderHeader
WHERE SalesPersonID IS NOT NULL
GROUP BY SalesPersonID, YEAR(OrderDate)) AS dt
```



Question 48

```
SELECT*
FROM Production.ProductPhoto
WHERE LargePhotoFileName LIKE '%green_%'
```

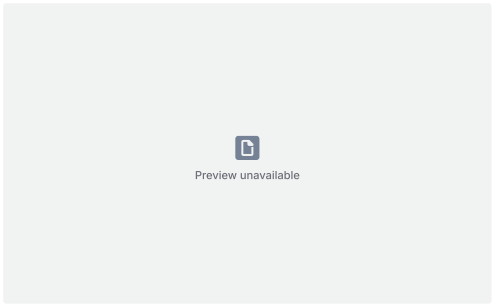


Question 49

```

SELECT pa.AddressLine1, pa.AddressLine2, pa.City,pa.PostalCode, sp.CountryRegionCode
FROM Person.BusinessEntityAddress ba
JOIN Person.BusinessEntity be
ON ba.BusinessEntityID = ba.BusinessEntityID
JOIN Person.Address pa
ON ba.AddressID = pa.AddressID
JOIN Person.StateProvince sp
ON pa.StateProvinceID = sp.StateProvinceID
```

```
10 WHERE sp.CountryRegionCode  NOT LIKE 'US'
```



Question 50

```
1 SELECT TOP (20) JobTitle, HireDate
2 FROM HumanResources.Employee
3 ORDER BY HireDate DESC
```

